JAVASCRIPT ASSIGNMENT : 07

Module 7: Arrays – Multiple Assignment Questions

Section 1: Creating and Accessing Arrays

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1. Create array of student names and print all names

Create array with at least five student names and print each name using loop.
Hint: Access elements using index.

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2. Access first and last element of array

Create array of numbers and print first and last element.
Hint: Use index zero and length minus one.

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3. Replace specific element in array

Create array of colors and replace one color with another.
Hint: Use index to modify value.

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4. Find length of array

Create array and print total number of elements.
Hint: Use length property.

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5. Create mixed data type array

Create array containing string, number, boolean and object. Print each element type.
Hint: Use typeof inside loop.

Section 2: push, pop, shift, unshift

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6. Add element at end using push

Create array of fruits and add new fruit at end.
Hint: Use push method.

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7. Remove last element using pop

Create array and remove last element. Print updated array.
Hint: pop returns removed value.

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8. Add element at beginning using unshift

Create array and add new element at start.
Hint: Use unshift method.

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9. Remove first element using shift

Create array and remove first element.
Hint: shift changes index positions.

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10. Create mini cart system using push and pop

Create shopping cart array where items can be added and removed.
Hint: Use push to add and pop to remove.

Section 3: map Method

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11. Multiply each element by two using map

Create array of numbers and return new array with each number doubled.
Hint: map returns new array.

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12. Convert array of names to uppercase

Create array of names and convert all to uppercase using map.
Hint: Use string method inside map.

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13. Extract specific property from object array

Create array of student objects and return array of student names.
Hint: map over object array.

Section 4: filter Method

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14. Filter even numbers

Create array and return only even numbers.
Hint: Return condition inside filter.

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15. Filter students with marks above 60

Create student array with marks and filter students who scored above 60.
Hint: Use filter on object property.

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16. Remove duplicate values

Create array with duplicates and remove duplicates using filter logic.
Hint: Compare index positions.

Section 5: reduce Method

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17. Calculate total sum using reduce

Create array of numbers and calculate total sum.
Hint: Use accumulator parameter.

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18. Find maximum number using reduce

Create array and find largest number.
Hint: Compare accumulator with current value.

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19. Count occurrences of elements

Create array of words and count how many times each word appears.
Hint: Use object accumulator inside reduce.

Section 6: find and forEach

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20. Find first number greater than 50

Create array and find first number greater than 50.
Hint: Use find method.

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21. Find student by id

Create array of student objects and find student with specific id.
Hint: Use find with condition.

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22. Print all elements using forEach

Create array and print index and value using forEach.
Hint: forEach gives element and index.

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23. Calculate sum using forEach

Create array and calculate sum manually using forEach.
Hint: Use external variable.

Section 7: Real World Logical Questions

24. Create shopping cart total calculator

Create array of products with price and quantity. Calculate total bill using reduce.
Hint: Multiply price and quantity.

25. Create student result analyzer

Create array of student objects and calculate average marks.
Hint: Use reduce for total then divide by length.

26. Create expense tracker summary

Create expense array and calculate total expense and filter expenses above specific amount.
Hint: Combine filter and reduce.

27. Group numbers into even and odd arrays

Take array and create two separate arrays one for even and one for odd numbers.
Hint: Use filter twice.

28. Sort array and remove duplicates

Create numeric array, sort it and remove duplicates.
Hint: Combine sort and filter logic.

29. Create simple leaderboard system

Create player score array and find highest scorer and top three scores.
Hint: Use sort and slice.

30. Convert array of strings into single string

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Create array of words and join them into sentence.
Hint: Use join method.